微分几何习题参考解答*

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1 第五章一部分习题

1. (3). 曲面第一基本形式为

$$I = \frac{dudu + dvdv}{(c + u^2 + v^2)^2}.$$

(u, v) 是等温参数，$\lambda = \frac{1}{c+u^2+v^2}$，故 Gauss 曲率

$$\begin{aligned} K &= -\frac{1}{\lambda^2}(\frac{\partial^2}{\partial u^2} + \frac{\partial^2}{\partial v^2}) \ln \lambda \\ &= (c + u^2 + v^2)^2 \frac{4c}{(c + u^2 + v^2)^2} \\ &= 4c. \end{aligned}$$

□

2. 取 $\tilde{S}$ 上局部正交参数 (u, v)，则有

$$\tilde{I} = E(du)^2 + G(dv)^2,$$

那么，

$$I = \lambda \tilde{I} = \lambda E(du)^2 + \lambda G(dv)^2.$$

*彭家贵、陈卿著《微分几何》，高等教育出版社 2021 年 11 月第 2 版

正交参数下 Gauss 曲率：

$$\begin{aligned}K &= -\frac{1}{\sqrt{\lambda E\lambda G}}\Big(\big(\frac{(\sqrt{\lambda E})_v}{\sqrt{\lambda G}}\big)_v + \big(\frac{(\sqrt{\lambda G})_u}{\sqrt{\lambda E}}\big)_u\Big)\\ &= -\frac{1}{\lambda}\frac{1}{\sqrt{EG}}\Big(\big(\frac{(\sqrt{E})_v}{\sqrt{G}}\big)_v + \big(\frac{(\sqrt{G})_u}{\sqrt{E}}\big)_u\Big)\\ &= \frac{1}{\lambda}\tilde{K}.\end{aligned}$$

□

3. (1).

$$K_S = \frac{1}{ab(1+u^2+v^2)^2},$$
$$K_{\tilde{S}} = \frac{1}{\tilde{a}\tilde{b}(1+\tilde{u}^2+\tilde{v}^2)^2},$$

故 $ab = \tilde{a}\tilde{b}$ 时对应地有 $K_S(u,v) = K_{\tilde{S}}(u,v)$.

(2). 设 $a \neq b$，$T : S \to \tilde{S}$ 是等距变换，$T(u,v) = (\xi,\eta)$。

那么 $K_S(u,v) = K_{\tilde{S}}(\xi,\eta)$，考虑两端最小值得到：

$$ab = \tilde{a}\tilde{b},$$

由 Gauss 曲率相等得到

$$u^2+v^2 = \xi^2+\eta^2,$$

令 $(u,v) = r(\cos\theta,\sin\theta)$，$(\xi,\eta) = r(\cos\varphi,\sin\varphi)$，即有

$$T^*\frac{\partial}{\partial r} = \frac{\partial}{\partial r},$$
$$T^*\frac{\partial}{\partial\theta} = \varphi_\theta\frac{\partial}{\partial\varphi}.$$

等距变换下第一基本形式相等，得到 $I(\frac{\partial}{\partial r},\frac{\partial}{\partial\theta}) = \tilde{I}(T^*\frac{\partial}{\partial r},T^*\frac{\partial}{\partial\theta})$，其中

$$I/(drd\theta) = -(a^2-b^2)\sin 2\theta + r^2\frac{a^2-b^2}{2}\sin 2\theta + r^2(\frac{a^2-b^2}{2})^2\sin 4\theta.$$

得到微分方程

$$\begin{aligned}&-(a^2-b^2)\sin 2\theta + r^2\frac{a^2-b^2}{2}\sin 2\theta + r^2(\frac{a-b}{2})^2\sin 4\theta\\ =&\varphi_\theta\Big(-(\tilde{a}^2-\tilde{b}^2)\sin 2\varphi + r^2\frac{\tilde{a}^2-\tilde{b}^2}{2}\sin 2\varphi + r^2(\frac{\tilde{a}^2-\tilde{b}^2}{2})^2\sin 4\varphi\Big),\end{aligned}$$

固定 r，积分后分析两端最值，可得下面几种种情况至少其一成立：
(a) $\varphi = \theta,\ (\tilde{a}, \tilde{b}) = \pm(a, b)$,
(b) $\varphi = \theta + \frac{\pi}{2},\ (\tilde{a}, \tilde{b}) = \pm(b, a)$.

对于 $a = b$ 的情形，类似地可以得到 $(\tilde{a}, \tilde{b}) = \pm(a, b)$. □

5.

$$d\mathbf{r}_\alpha = (\Gamma^\gamma_{\alpha\beta}\mathbf{r}_\gamma + b_{\alpha\beta}\mathbf{n})du^\beta,$$
$$D\mathbf{r}_\alpha = \Gamma^\gamma_{\alpha\beta}\mathbf{r}_\gamma du^\beta,$$

$$\begin{aligned} D\mathbf{v} &= D(f^\alpha \mathbf{r}_\alpha) \\ &= df^\alpha \mathbf{r}_\alpha + f^\alpha D\mathbf{r}_\alpha \\ &= (df^\alpha + \Gamma^\alpha_{\beta\gamma} f^\beta du^\gamma)\mathbf{r}_\alpha. \end{aligned}$$

□